

To: Dr. Haonan Lin

From: Mary Martin, Isra Hussain, John Garrett, Mackenzie Davis, and Andrea Creel

CC: Chikamnele Nkwocha, Ethan Baird

Subject: BMED 2250 A03 Team #4 Technical Progress Report #3

Title: Effects of Extreme Temperature on Insulin Administration

Date: March 18 – April 28, 2026

Executive Summary:

During Phase 3, our team successfully developed a functional prototype of a cooling sleeve fan system designed to regulate the temperature of insulin pumps in extreme heat environments. The prototype integrates an active fan-based cooling component with a compression UV sleeve to help maintain insulin within its safe operating threshold ($\leq 30^{\circ}\text{C}$), improving reliability and patient safety in extreme heat environments.

Our primary user group consists of low-income individuals with Type 1 diabetes living in rural Georgia who experience energy insecurity and prolonged exposure to extreme heat. These conditions increase the risk of insulin degradation, which can be extremely harmful to patients with diabetes.

During this phase, we explored two design ideas. The first involved a chemically activated cooling gel system intended to regulate temperature across the sleeve without constant refrigeration. This approach proved to be too complex to prototype due to formulation challenges and limited feasibility within project constraints. As a result, we chose to prototype and test our second design idea, which was an active fan-based cooling system. Even though the design and final test plan were simple, we successfully prototyped and demonstrated the ability to reduce temperature around the insulin pump, validating its potential as a practical solution.

Moving forward, we could refine the Arduino-based system by optimizing sensor integration and developing a more compact and comfortable layout for the circuit board to sit on the arm.

Progress Summary:

During Phase 3, our group made significant progress in developing our final prototypes, both mathematical and physical. We decided to create a cooling sleeve based on our user needs including affordability, effectiveness, and ease of use, we narrowed down our device design concepts to either an automated cooling system with the use of fans or a manual cooling system with the use of a gel. Based on the scope of this class and the need for proof of concept with the Arduino and breadboard, we decided on the fan idea. We wrote code that turned on an LED light and a fan when temperatures were detected above a certain threshold and created a prototype with an affordable, breathable fabric that had this concept attached to it. Within phase 3, our problem statement, user population and needs, and design inputs/constraints stayed the same.

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Problem Statement:

Lower-income individuals living in environments that experience extreme heat may not have access to climate-controlled housing, creating significant issues in the administration of insulin in diabetics. Automated Insulin Delivery systems are not designed to reliably function outside of standard temperature ranges. Therefore, lower-income individuals need a way to properly keep insulin temperatures regulated for effective administration.

Core Physiology:

When discussing physiological information, it is important to understand the biological systems connecting to insulin pumps to analyze the project at hand. Insulin, as a component, is temperature-sensitive and degrades exponentially when exposed to extreme heat environments. When discussing biological systems connecting with insulin in the body, the compound travels in the body with regulation in organs before degradation with synthesis and secretion by pancreatic β -cells, followed by clearance, delivery, action on muscle, and final degradation (Tokarz et al., 2018). After this, insulin is created as a precursor molecule via folding and then travels to the liver for degradation (Tokarz et al., 2018). To apply this to extreme hot temperatures that cause problems in regulation, insulin usually degrades with cell-mediated organs, yet temperature stress can cause problems in folding in the insulin proteins, causing an inability for receptors to bind. Thus, according to Victoria Tokarz and her associates, elevated glucose levels due to an inability for GLUT4 to be activated creates glucose uptake where insulin fails to signal cells. Connecting to our current project in insulin degradation issues with pumps, thermal stress causes stabilizing interactions, such as three-dimensional components to inhibit the receptor binding and activity, which causes a decrease in aggregation rates and lowers the impact of glucose, creating lower insulin (Azam et al., 2022).

User Research:

Our user is a lower-income individual with type I diabetes living in a rural community. This population requires a diabetes management solution that protects insulin in extremely hot environments while remaining affordable, portable, and simple to use. Our research during each of the three phases focused on lower-income individuals, rural populations, and people experiencing housing and energy insecurity. We learned that lower-income individuals experience higher rates of diabetes and poorer disease management outcomes due to limited access to healthcare, medical supplies, and stable living conditions (Centers for Disease Control and Prevention, 2023). Furthermore, rural populations have a higher prevalence of diabetes compared to urban populations because of limited access to healthcare services and climate-

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controlled housing (O'conner et al., 2025). People without climate-controlled housing are less likely to have reliable temperature control, increasing the health risk for chronic conditions such as diabetes that require temperature-sensitive medication (Saelee et al., 2024).

Furthermore, we learned from research that people with diabetes have impaired thermoregulation due to vascular and nerve damage (Centers for Disease Control and Prevention, 2025). Impaired thermoregulation reduces the body's ability to dissipate heat, which increases susceptibility to dehydration and metabolic instability during extreme heat. Since insulin degrades at high temperatures, our solution involves a cheap and portable cooling system that maintains insulin within a safe temperature range. Specifically, we decided to create a cooling sleeve that goes over an insulin patch, using automatic fans to ensure that the patch remains within the upper limit of the standard temperature range so that insulin does not degrade. We created the sleeve out of affordable materials to be accessible to our population, and the fans and microcontroller used were also affordable, effective, and easy to use.

Results of Ideation:

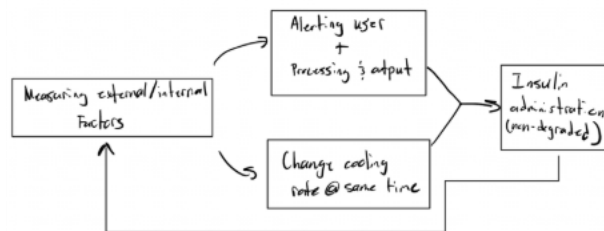
Our group started with a total of ten designs in phase 2 before we narrowed our focus down to the cooling sleeve. We felt like this idea best met the needs of our user group due to its overall low-tech design, ease of use, and effectiveness in keeping insulin from degradation in hotter temperatures. The physical design of the device is based off compression sleeves, with the sleeve being made primarily of an elastic fabric to fit snugly around the user's arm. We also included an insert for the insulin pump to slot into place, ensuring that the user's movement will not shift the pump and cause discomfort. Our initial ideation was to have the inner linings of the insulin pump composed of cooling gel and insulating fabric to keep the pump within the temperature range without causing discomfort to the user due to cold shock. The main issue we faced was with the gel, due to the gel used in the Frio insulin wallet being copyrighted and the chemical processes to make the gel automatically cool. At the start of phase 3, we came up with two potential design ideas. The first involved a chemically activated cooling gel system intended to regulate temperature across the sleeve. The gel would be in the lining of the cooling sleeve and regulate temperature similarly to our initial ideas in phase 2. However, this gel approach proved to be too complex to prototype due to formulation challenges and limited time. As a result, we chose to prototype and test our second design idea, which was an active fan-based cooling system. We decided to write code that turned on an LED light and a fan when temperatures were detected above a certain threshold. We planned to attach the microcontroller and mini fan to a compression sleeve on the arm as an effective cooling mechanism.

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Project Changes:

Since the Phase 2 report, our project has become extremely focused and technical. In Phase 3, we narrowed our focus down to the cooling sleeve and specifically decided on the use of fans for automatic cooling. Our process shifted from the propagation of many design ideas in phase 2, where we had many different potential design concepts, to creating a feasible mathematical and physical prototype. In phase 3, we developed code with logic to turn on LED lights to alert users when above a certain threshold (where insulin degrades when it is too hot), and then helped with this alert by activating a cooling mechanism through fans to promptly begin addressing the problem we set out to solve in phase 1. While in phase 2, there were only theoretical solutions, phase 3 gave us the opportunity to actively develop a solution within the scope of this class that met our user needs and intended design constraints/inputs.

Results of Testing:

The primary objective of our testing phase was to ensure the effectiveness of the active fan-based cooling sleeve in maintaining a temperature below the critical insulin degradation threshold of 30°C. Due to a faulty temperature sensor, our team manually input temperature values into the Arduino code to verify the prototype's control logic. We confirmed that the system successfully triggered both the LED visual alert and the mini-fan when a value greater than or equal to 30°C was inputted. While we implemented a 2°C buffer zone in the code to keep the cooling mechanism active until the temperature dropped below 28°C, the faulty sensor meant we had to verify this logic through manual code adjustments rather than real-time thermal changes. Beyond the software logic, the physical prototype demonstrated successful hardware integration, with the mini-fan providing consistent airflow and the breathable compression fabric maintaining a lightweight, portable profile. However, future iterations will require a design input update to include a more robust thermistor or a multi-sensor array to ensure reliable, real-time automation. While the current prototype successfully addresses the user needs for affordability and portability, a future constraint regarding power efficiency should be added to ensure the fan can

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operate for extended periods in rural areas with limited energy access. Ultimately, these results demonstrate that the active fan-based cooling system is a viable solution for protecting insulin from thermal stress and preventing the physiological degradation of the medication.

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References:

Azam, M. S., Wahiduzzaman, M., Reyad-ul-Ferdous, M., Islam, M. N., & Roy, M. (2022). Inhibition of insulin degrading enzyme to control diabetes mellitus and its applications on some other chronic disease: a critical review. *Pharmaceutical Research*, 39(4), 611-629.

Centers for Disease Control and Prevention. (2025). Rural–urban disparities in state-level diabetes prevalence among U.S. adults, 2021. *Preventing Chronic Disease*, 22.

https://www.cdc.gov/pcd/issues/2025/24_0199.htm.

O'Connor, A., Wellenius, G., & Rural Health Research Center. (2025). Rural–urban disparities in state-level diabetes prevalence among U.S. adults, 2021. *Preventing Chronic Disease*, 22.

Tokarz, V. L., MacDonald, P. E., & Klip, A. (2018). The cell biology of systemic insulin function. *Journal of Cell Biology*, 217(7), 2273-2289.

Appendix A: Arduino Code

```
#include <Arduino.h>
const int FAN_PIN = 3;
const int LED_PIN = 9;
// — Insulin Protection Thresholds —————
const float TEMP_THRESHOLD_HIGH = 30.0; // Fan ON
const float TEMP_THRESHOLD_LOW = 28.0; // Fan OFF
bool isFanOn = false;
void setup() {
  pinMode(FAN_PIN, OUTPUT);
  pinMode(LED_PIN, OUTPUT);
  Serial.begin(9600);

  Serial.println("--- Testing Mode: Hard-Coded Temperature ---");
}
void loop() {
  // — MANUALLY CHANGE THIS VALUE TO TEST —————
  float tempC = 45.0; // <--- Change this number to 25.0, 30.0, etc.
  // —————
  // Logic: Turn ON at 30, stay ON until it drops below 28
```

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```
if (tempC >= TEMP_THRESHOLD_HIGH) {
    isFanOn = true;
} else if (tempC <= TEMP_THRESHOLD_LOW) {
    isFanOn = false;
}
// Update hardware
digitalWrite(FAN_PIN, isFanOn ? HIGH : LOW);
digitalWrite(LED_PIN, isFanOn ? HIGH : LOW);
// Status Printout
Serial.print("Test Temperature: ");
Serial.print(tempC);
Serial.print(" C | Fan Status: ");
Serial.println(isFanOn ? "ON (Cooling)" : "OFF (Safe)");
delay(2000); // Slow down the loop for easier reading
}
```

Appendix B: Design Documentation Spreadsheet

You should NEVER have to edit these columns - it will autopopulate				You will edit this column - when told to by a facilitator or TA	
Problem Statement					
Current Iteration		PS.3		Approvals	
Problem Statement (Version PS.3)		Lower-income individuals living in environments that experience extreme heat may not have access to climate-controlled housing, creating significant issues in the administration of insulin in diabetics. Automated Insulin Delivery systems are not designed to reliably function outside of standard temperature ranges. Therefore, lower-income individuals need a way to properly keep insulin temperatures regulated for effective administration.		Prof. Haonan Li. Approved on 3/17/2026	
User Needs					
User Need	Core	Text	Inputs	Approvals	
UN.1	Yes	Affordable	3	Our user needs were all approved on 3/17 following the phase II presentation.	
UN.2	Yes	Reliable	3		
UN.3	Yes	Easy to Use	2		
Design Inputs					
Design Input	Linked UN	Text	Iteration	Approvals	
DI.1	UN.2	Maintain insulin temperature below 30° Celsius at extreme temperatures	1	Our design inputs 1-6 were approved on 3/17.	
DI.2	UN.1	Low cost for accessibility	1		
DI.3	UN.4	Compatible with existing insulin pumps without extensive modifications	1		
DI.4	UN.4	Lightweight and compact (<200g)	1	The other 4 design inputs were approved on 4/12. (see DI.10 in other tab, template is not working property).	
DI.5	UN.3	Quick activation	1		
DI.6	UN.1	Passive/Low-Power	1		
DI.7	UN.2	(when above 30 degree Celsius threshold)	1		
DI.8	UN.3	Simple visual status indicator (LED or color indicator)	1		
DI.9	UN.1	power sources	1		
DI.10					

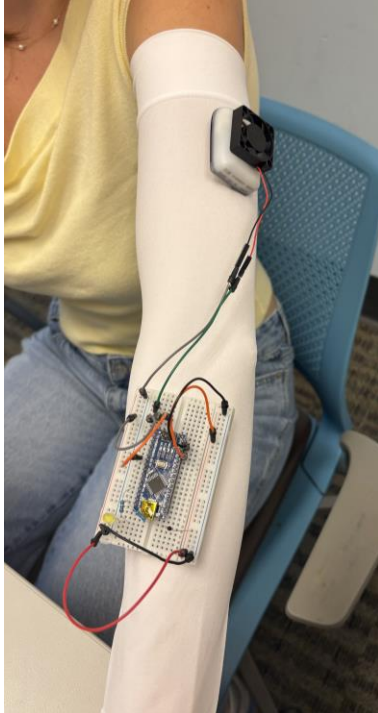
Appendix C: Physical Prototype and Evaluation Criteria

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Evaluation Criteria:

Device	Affordability	Effectiveness	Ease of Use	Total	Rank
Fan Design	7	7	8	22	1
Gel Design	5	8	5	18	2